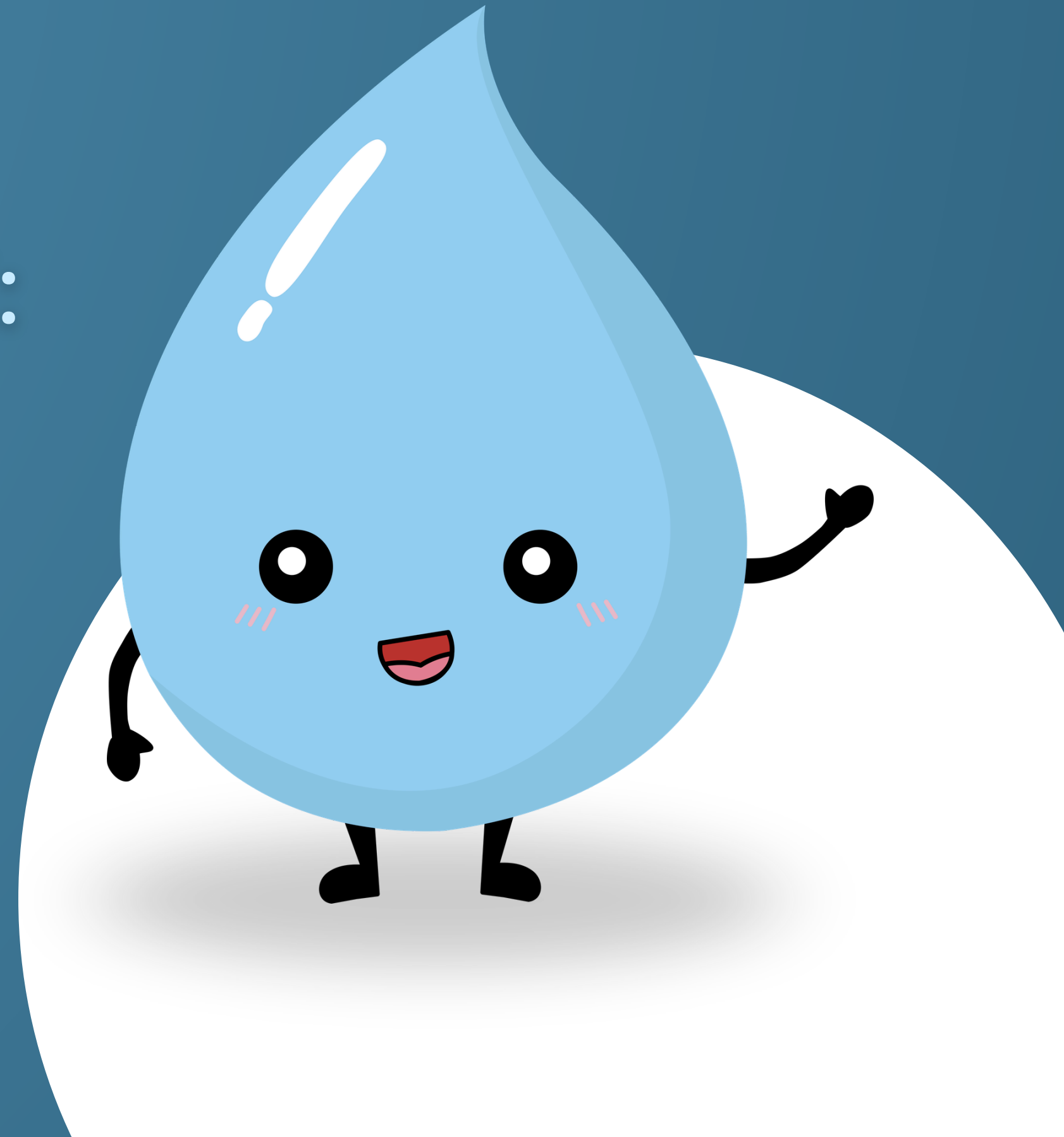
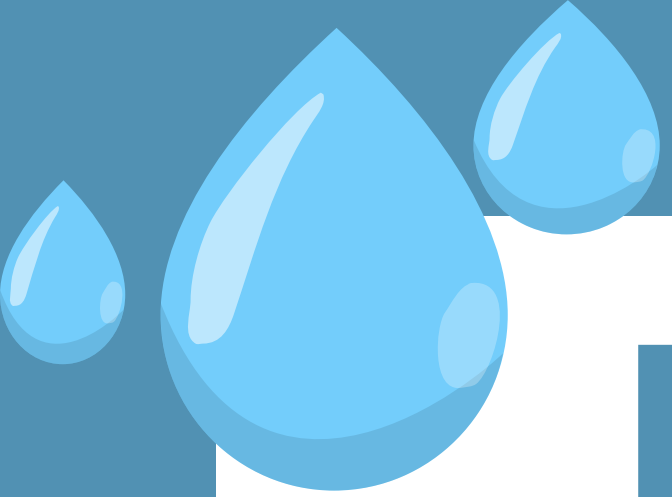


FROM PIPES TO PEOPLE: **ADDRESSING LEAD** IN DRINKING WATER

Presented by:
UWO TEAM HS 3240B





PROJECT OVERVIEW

- Our **health-based action toolkit** is a resource for stakeholders and the public to understand **lead contamination in drinking water** and support effective prevention strategies.
- It compiles key research into **clear, accessible, and actionable** information for diverse users, including families, youth, community organizations, and decision-makers.
- The toolkit includes original deliverables such as **infographics, youth-friendly materials, databases**, and a **cartoon PSA**, along with planning tools like a **logic model, stakeholder matrix**, and **DPSSEA model**.
- By combining research with practical resources, the toolkit builds **awareness, fosters collaboration**, and **empowers informed action**.



DELIVERABLES

INFOGRAPHICS



Learning to be a WATER DETECTIVE

The **LEAD** Method to Find Lead in Water

Lead in water is tricky because you **can't see it, smell it, or taste it.**¹ That's why detectives use clues!

L Look For Clues 

- are the pipes or faucets very old?
- does the water sometimes look **brown or cloudy**?¹

E Evaluate Risk 

- ask an adult when the **pipes** were last replaced
- homes built before **1986** may contain lead¹

A Analyze Water 

- use a water test kit
- send water to a **lab** for testing¹

D Decide What To Do 

- run **cold** tap water for 30 seconds before drinking¹
- use a water **filter**

Lead can make you **sick**, especially your **brain** and body while growing.¹



LEAD IN WATER: A PARENT'S GUIDE

Lead can enter drinking water from old plumbing and can **harm** children's brain development.

If you **suspect** lead contamination:

➔ NEXT STEPS

TEST YOUR WATER
Your **local water authority** is always your **first** source for testing and identifying lead contamination in your tap water.¹

INQUIRE
Ask your water provider if there is a **lead service line** providing water to your home.
• Individual risks to lead exposure increase while the work is ongoing and for up to six months after the work is completed.¹

REPLACEMENT
If your home has a lead service line, the solution is replacement.
• Check with your municipality or water utility about programs or incentives to help with replacement.¹

FILTRATION
Using a filter can reduce lead in drinking water.
• If you use a filter, make sure you use a filter **certified** to remove lead.²

MEDICAL ATTENTION
A family doctor or pediatrician can perform a blood test for lead and provide information about the health effects of lead.²

WHO TO CONTACT

YOUR CHILD'S DAYCARE
The faucets that provide water used for consumption, including drinking, cooking lunch, and preparing juice and infant formula, should be tested.²

LOCAL PUBLIC HEALTH UNIT
Municipalities typically provide **testing kits**, advise flushing water lines, coordinate public-side lead pipe **replacements**, and sometimes offer subsidies or **loans** for private replacements.⁴

Lead **cannot** be detected by the senses. The **only way to know if your water has lead is to test it**¹ and reach out for help.

Concerned about lead in your drinking water?
Scan the QR code to access the parent resource database.

SCAN ME



LEAD IN YOUR WATER

About Lead¹

- Metal found naturally and artificially in the environment.
- **Toxic metal**
- **NO** known safe level of exposure
- Found in tap water from lead plumbing

Canadian Guidelines

0.005 mg/L maximum amount of lead in water¹

Lead is common in...

Homes built before 1952

Be aware of...
Brass faucets and fitting
Lead solder used in plumbing
Lead service lines from water main¹

1.5 million deaths globally were attributed to lead exposure³

33 million years lost to disability³

AVOID: Hot water as it accelerates lead leaching into water¹

Health Effects

- Cognitive and behavioural issues.²
- Increased blood pressure.²
- Kidney dysfunction.³
- Cardiovascular diseases.²
- Liver damage.³
- Immunotoxicity and toxicity to reproductive organs.³
- Lead is stored and accumulates in bones and teeth.³

Those at risk²

- Infants & Children
- Fetuses
- Pregnant & Menopausal women
- Low-income or minority communities

Children can absorb 4-5x as much lead as adults from an ingested dose³

What to look for

- **Greyish-black** water service lines¹
- **Soft or easily dented** when scraped with a knife¹

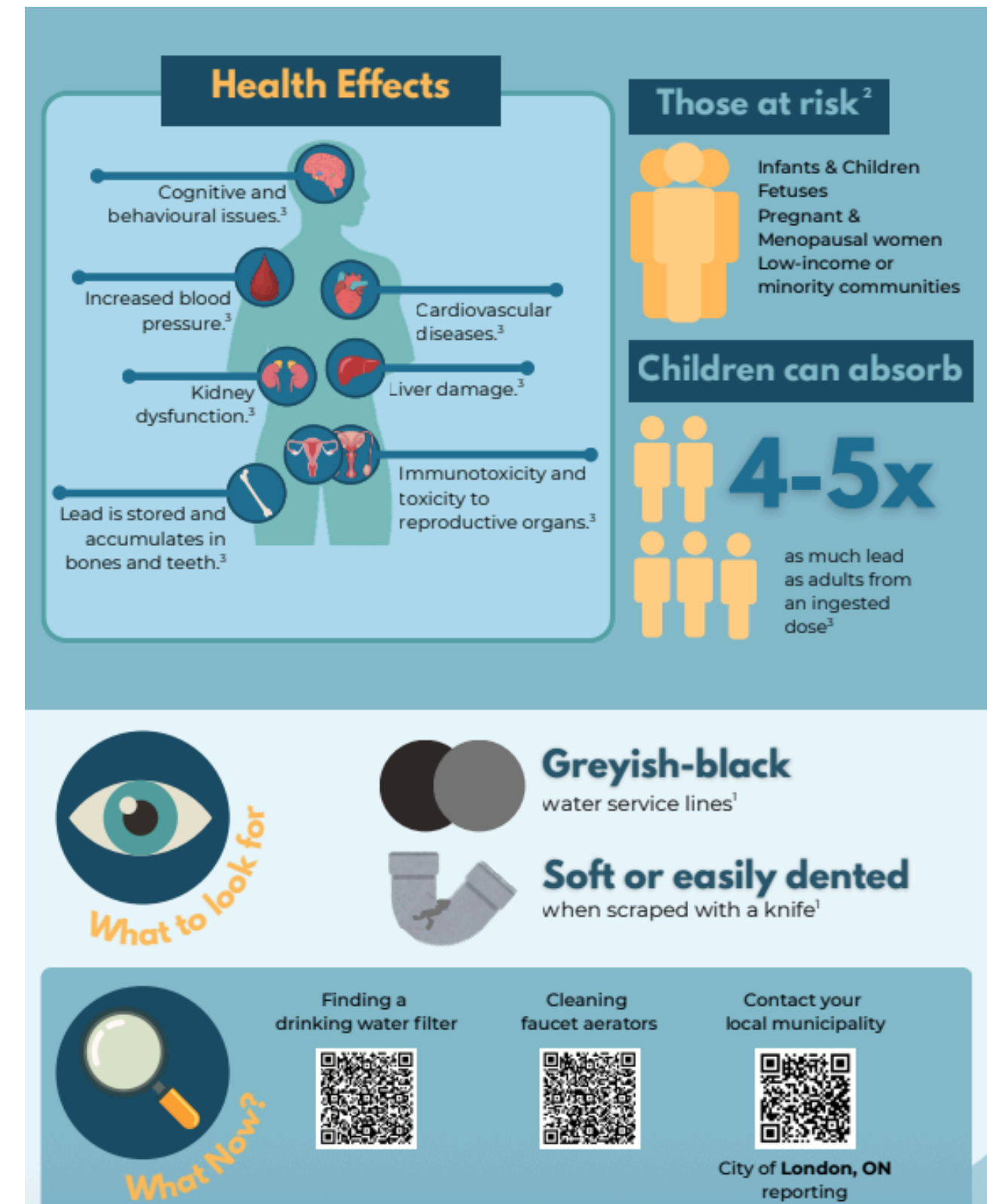
What Next?

- Finding a drinking water filter
- Cleaning faucet aerators
- Contact your local municipality

City of London, ON reporting



INFOGRAPHICS



INFOGRAPHICS



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SCAN ME

GLOCAL FOUNDATION OF CANADA

Credits for: Disease Control and Prevention. (2023, August 29). About lead in drinking water. <https://www.cdc.gov/lead/prevention/prevention/drinking-water.html>
Environment Canada. (2018, October 15). Lead in drinking water: What you need to know. <https://www.ec.gc.ca/lead-in-drinking-water/what-you-need-to-know>
Health Canada. (2018, October 15). Drinking water: What you need to know. <https://www.canada.ca/en/health-canada/services/drinking-water/quality-of-drinking-water/lead-in-drinking-water.html>
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DELIVERABLES

CARTOON VIDEO



DELIVERABLES

REPORTS

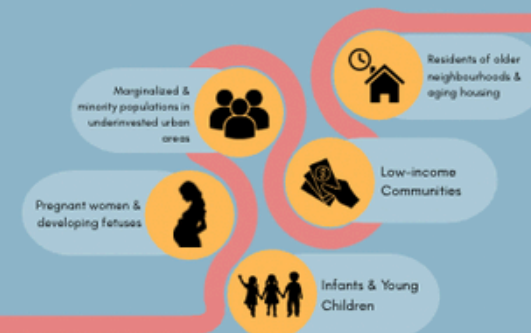
Toxicological Report: Lead (Pb)

Lead (Pb) is a highly poisonous and non-biodegradable pollutant that persists and bioaccumulates within the human body over time, with risks increasing the longer someone is exposed. The toxicity of this metal is dependent on how much is absorbed, the route of entry, and the duration of exposure (Sani & Amanabo, 2021). Human exposure to Pb is mainly attributable to two primary media, the air and surfaces, entering the body via inhalation, ingestion, transdermal, and percutaneous approaches.

Though occupational settings are the main drivers of Pb inhalation, the general population is primarily exposed through contaminated drinking water via the oral route (Sani & Amanabo, 2021; Agency for Toxic Substances and Disease Registry [ATSDR], 2020). Children are especially vulnerable due to their innate hand-to-mouth, or object-to-mouth behaviours, as well as their higher gastrointestinal absorption rates compared to adults (Sani & Amanabo, 2021).

Epidemiological Report: Who is at Risk?

Lead (Pb) contamination remains a critical public health crisis, as well as a silent environmental justice emergency. Over time, general exposure to Pb in tap water has declined due to legislative actions; however, legacy contamination within aging housing and water infrastructure continues to endanger people. Contamination from plumbing materials is noted to still be a significant contributor to Pb poisoning in many high-income countries today (Yu, 2025). Pb primarily enters drinking water through corrosion, which describes a chemical reaction where Pb pipes, faucets, and plumbing fixtures wear away into the water supply that people utilize and consume. Corrosion is more severe when the water has a high acidity or low mineral content. Since Pb is non-biodegradable, it remains persistent in the environment and bioaccumulates in the body over time, leading to devastating health effects even at low exposure levels (Centers for Disease Control and Prevention [CDC], 2025).



Infants and children represent the most vulnerable populations due to their rapid

Risk Assessment Framework

Hazard Identification =

Lead (Pb) is a toxic heavy metal that poses significant risks to human health even at low exposure levels (Sani & Amanabo, 2021). Contamination in drinking water primarily occurs through the corrosion of Pb service lines, plumbing fixtures, and solder used in older infrastructure (Agency for Toxic Substances and Disease Registry [ATSDR], 2020). It is a multi-organ toxin affecting nearly every system in the body, specifically the neurological, hematological, cardiovascular, and renal systems (Sani & Amanabo, 2021).

Target Organs =

Once Pb enters the body, it is distributed mainly to the brain, liver, and kidneys, with the majority (94% in adults; 76% in children) being stored in the teeth and bones of the skeletal system (ATSDR, 2020).

The primary hazard is permanent neurodevelopment damage, including intelligence quotient (IQ), shortened attention spans, and increased antisocial behaviours. In adults, chronic hazards include hypertension, increased risk of heart disease, anaemia, and chronic kidney disease (CKD) (Sani & Amanabo, 2021).

Dose Assessment =

A key finding in Pb toxicology is that no safe blood Pb level (BLL) has been identified for any age group (World Health Organization [WHO], 2023).

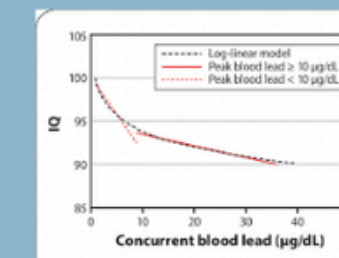


Figure 1.1: The supra-linear, dose-response curve between BLL and IQ (Matthew et al., 2014).

The supra-linear relationship between the magnitude of Pb dosage and the body's response to it. The dose-response curve for cognitive impairment is supra-linear, meaning that the steepest decline in IQ occurs at the lowest levels of exposure.

DELIVERABLES

STAKEHOLDER MATRIX



Stakeholder Register

High Influence & Low Interest

MECP

- www.ontario.ca/page/drinking-water
- R243LeadSelf-Report@ontario.ca
- (855) 515-1331

High Influence & High Interest

Water Customer Service

- water@london.ca
- (519) 661-4739

Lead Testing

- leadtesting@london.ca
- (519) 661-2489 x 5541

Low Influence & High Interest

Thames Valley District School Board

- (519) 452-2000

London District Catholic School Board

- (519) 663-2088

London Plumbing

- 24hr Plumbing Emergency Services
- (226) 270-0405

Plumbing For Life

- info@plumbingforlife.ca
- (519) 860-0219

Stakeholder Analysis Matrix

Satisfy

Provincial Government - Ontario

- Ministry of the Environment, Conservation and Parks (MECP) regulates lead contamination under the Safe Drinking Water Act, 2002.
- Drinking water quality standard (maximum 10 µg/L for lead; Ontario Regulation 169/03).
- Schools and child care centers are required to flush their plumbing in accordance to Ontario Regulation 243/07

Engage

Municipal Government - London

- Operates the municipal drinking water system and maintains water mains and pipe systems.
- Conducts lead testing services and implements public lead service line replacements.

Public Health Units - Middlesex-London Health Unit

- Monitors communities health risk and provides education on lead exposure.
- Collaborates with municipalities to ensure provincial standards.

INTEREST

General Public

- Businesses not dependent on municipal water
- Mass media shifts in public attention or concern

Communities - Homeowners, Residents, Schools, Plumbers and Contractors

- Directly affected and receives information on testing and risks.
- Participates in water testing and pipe replacement programs.
- Contractors/Plumbing industry replaces lead service lines and plumbing systems.

Monitor

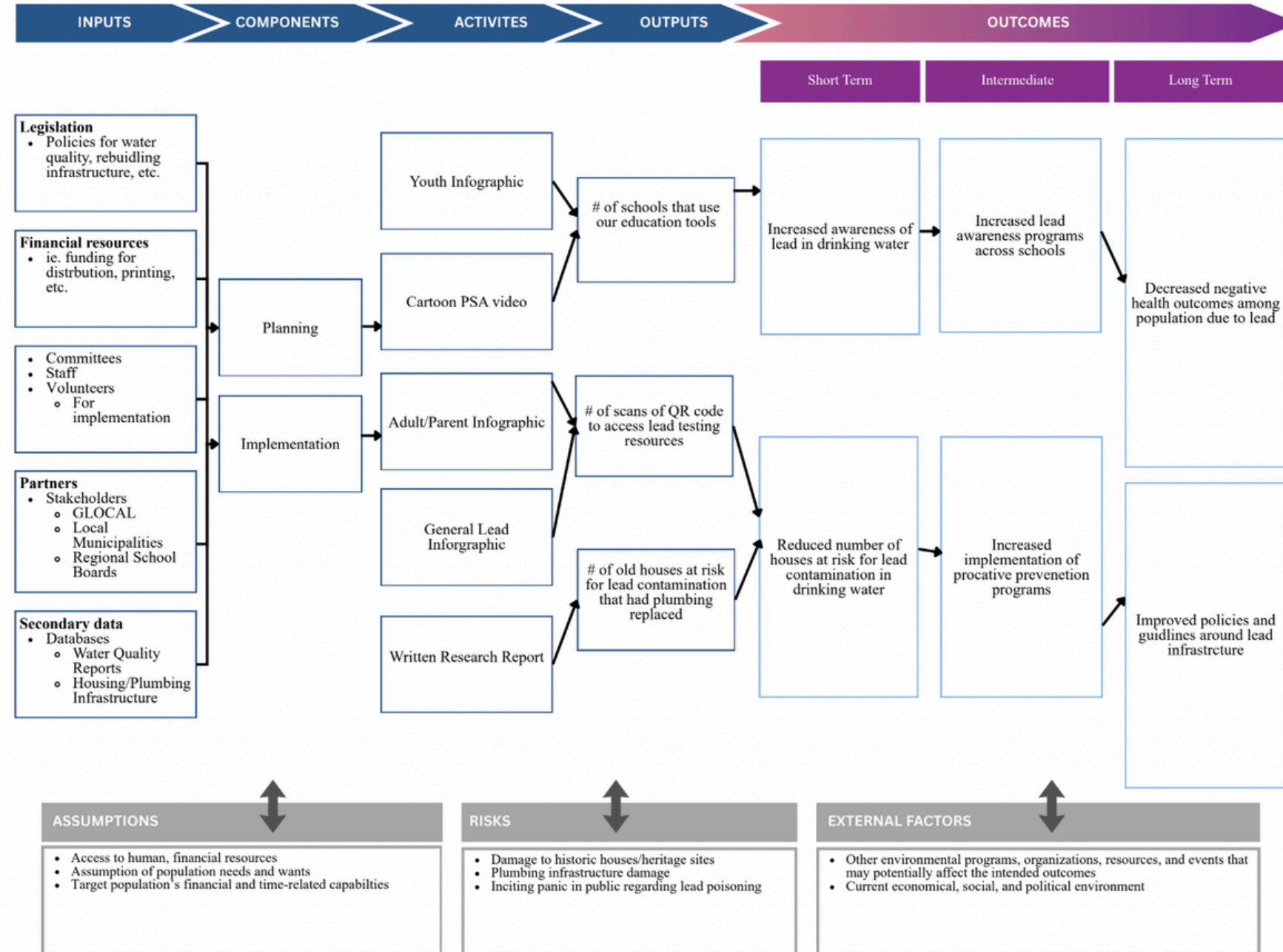
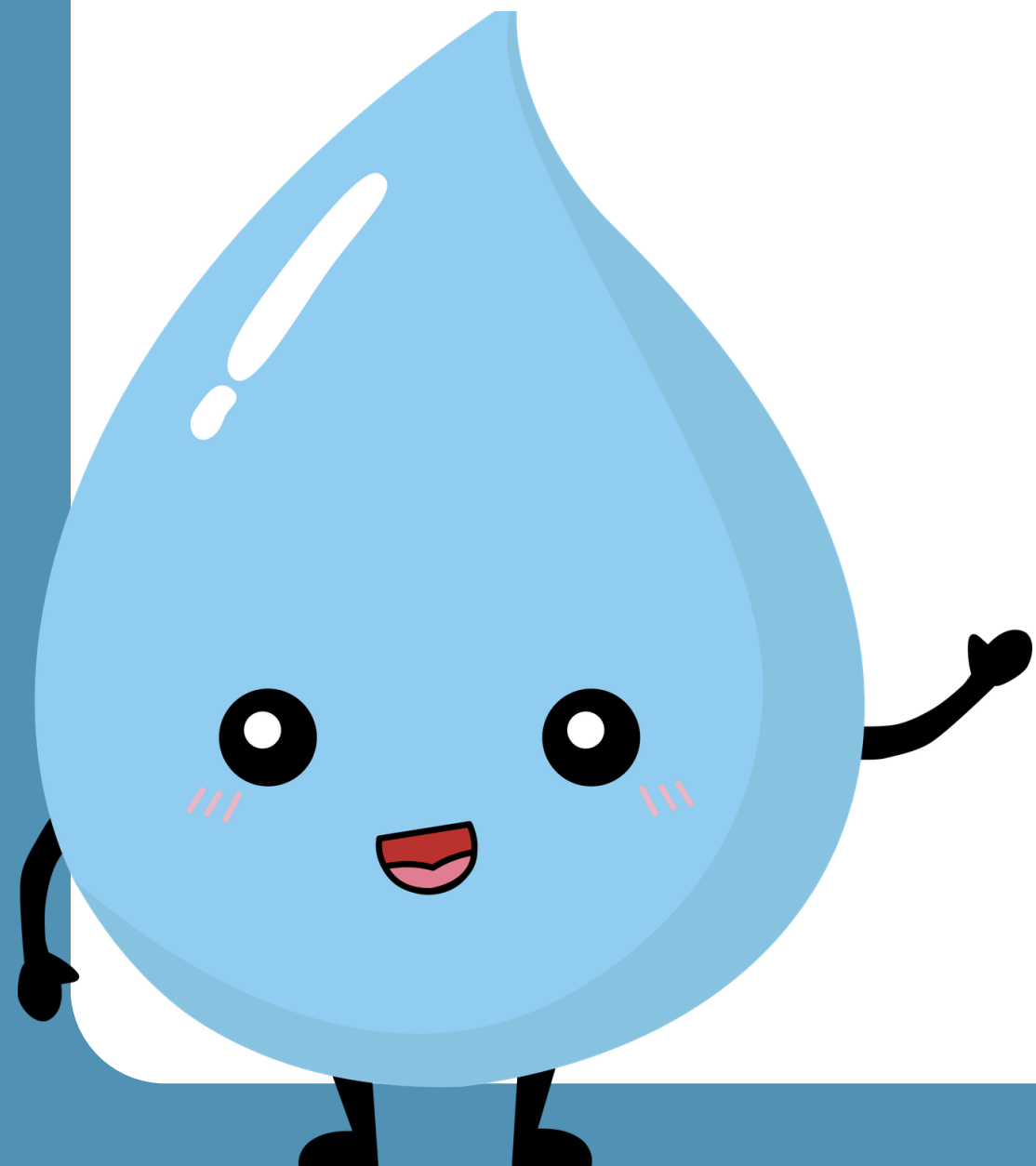
INFLUENCE

Inform



DELIVERABLES

PROGRAM LOGIC MODEL



DELIVERABLES

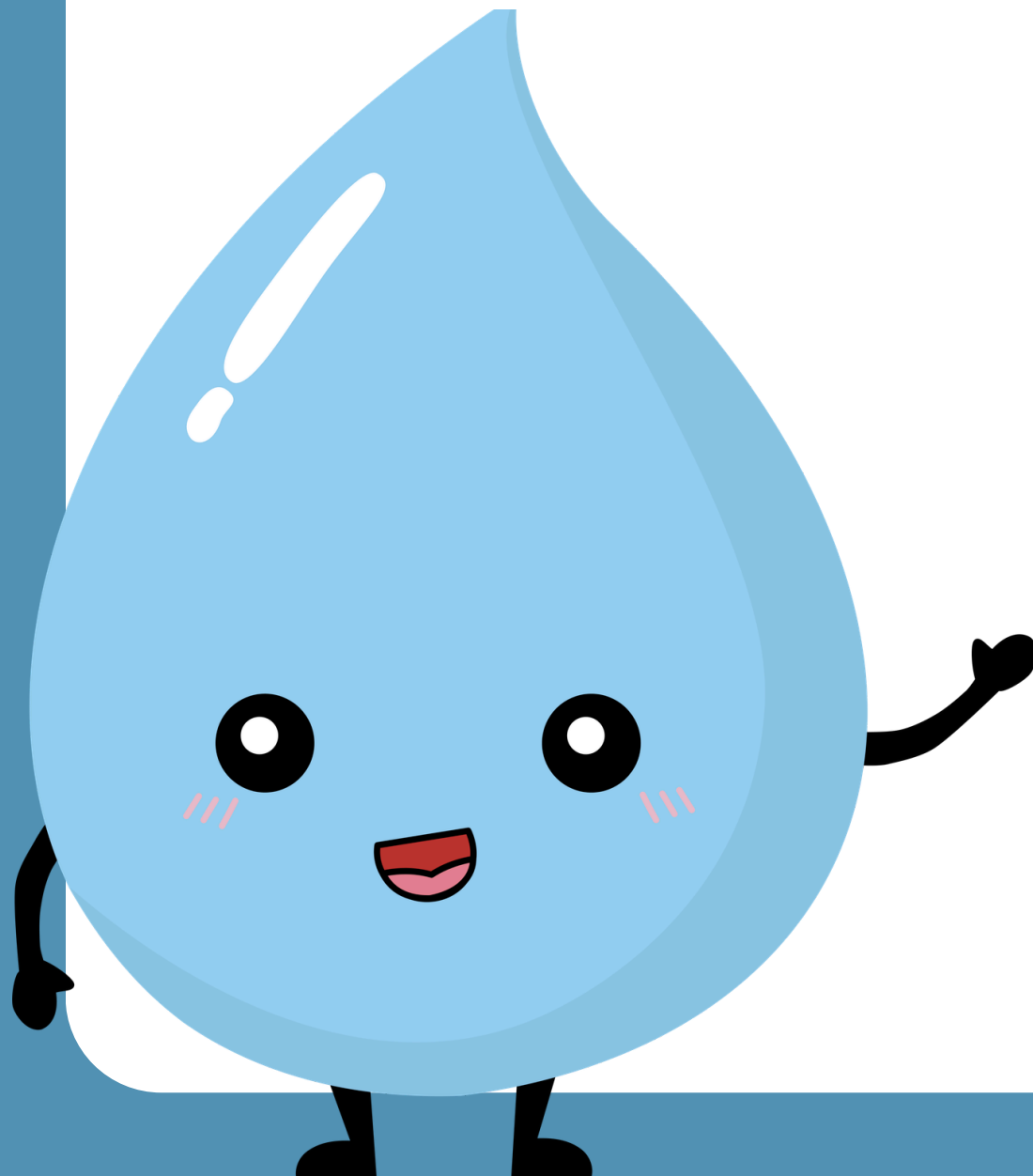
DPSSEA MODEL

Driving Forces

- Aging housing and plumbing infrastructure in London, especially in homes built before 1952 or with pre-1975 plumbing
- Long-term use of lead-containing service lines, lead solder, and brass fixtures
- Unequal housing conditions, especially in older central neighbourhoods and lower-income rental housing
- Limited awareness among residents about where lead comes from, how exposure happens, and who is most at risk
- Structural inequities that make it harder for some households to afford plumbing upgrades, filters, or testing

Pressures

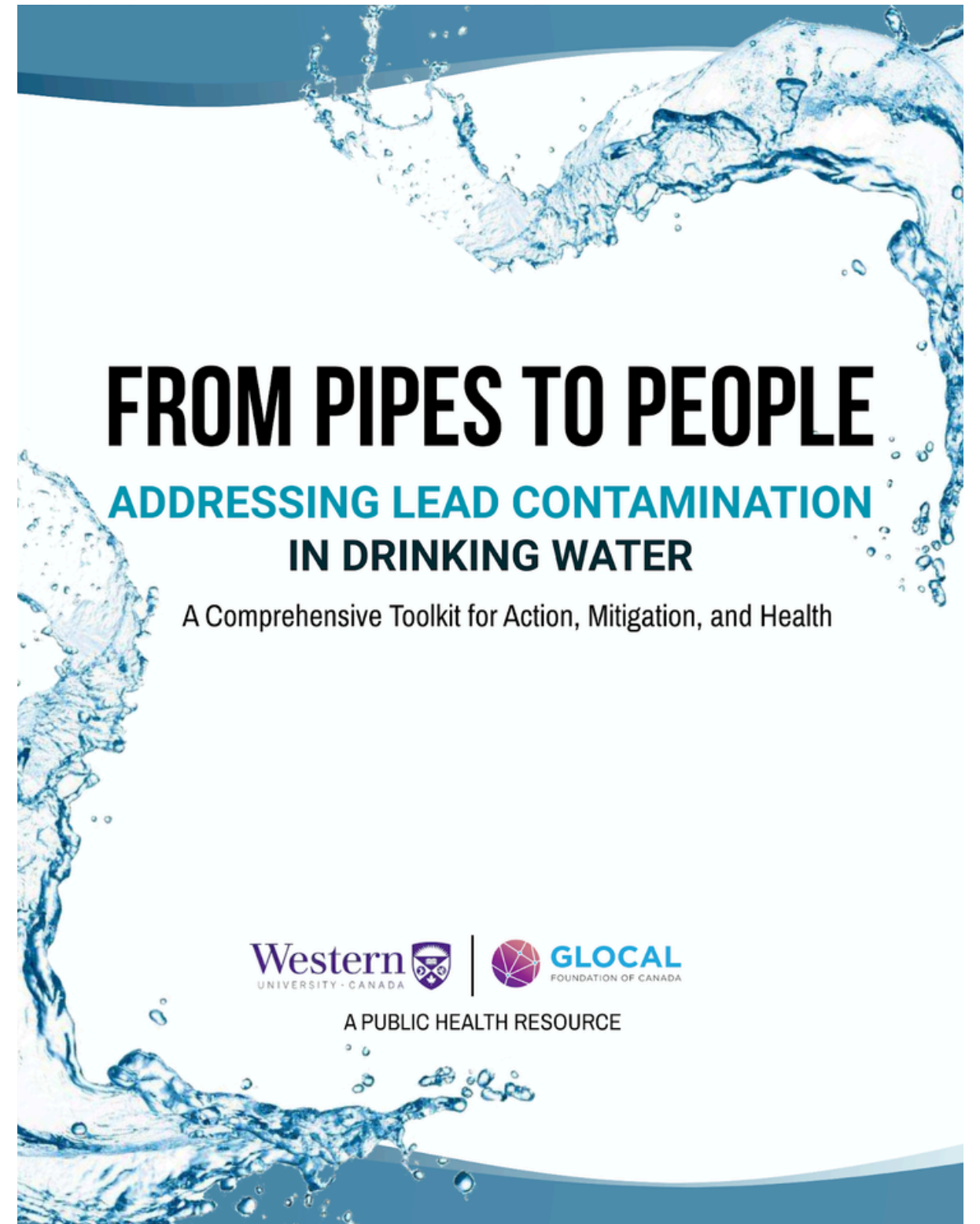
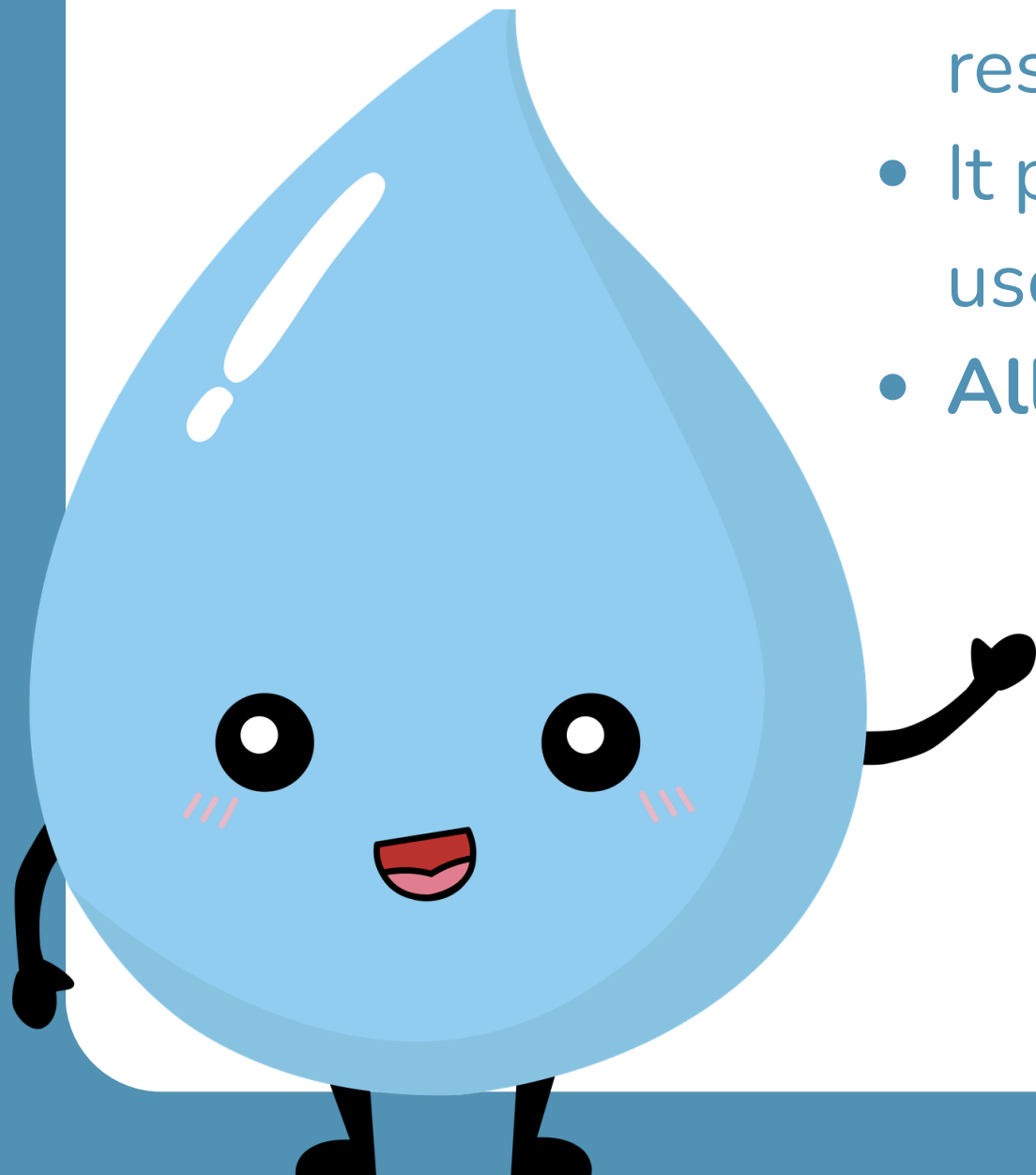
- Lead leaching from old lead service lines into household tap water
- Lead particles breaking off from pipes, solder, and fixtures
- Water sitting stagnant in pipes for several hours, increasing leaching
- Use of hot tap water, which accelerates lead release
- Incomplete replacement of only part of a lead service line, which may not fully solve the problem
- Limited sampling at a small number of municipal locations, which may not reflect exposure inside older private homes



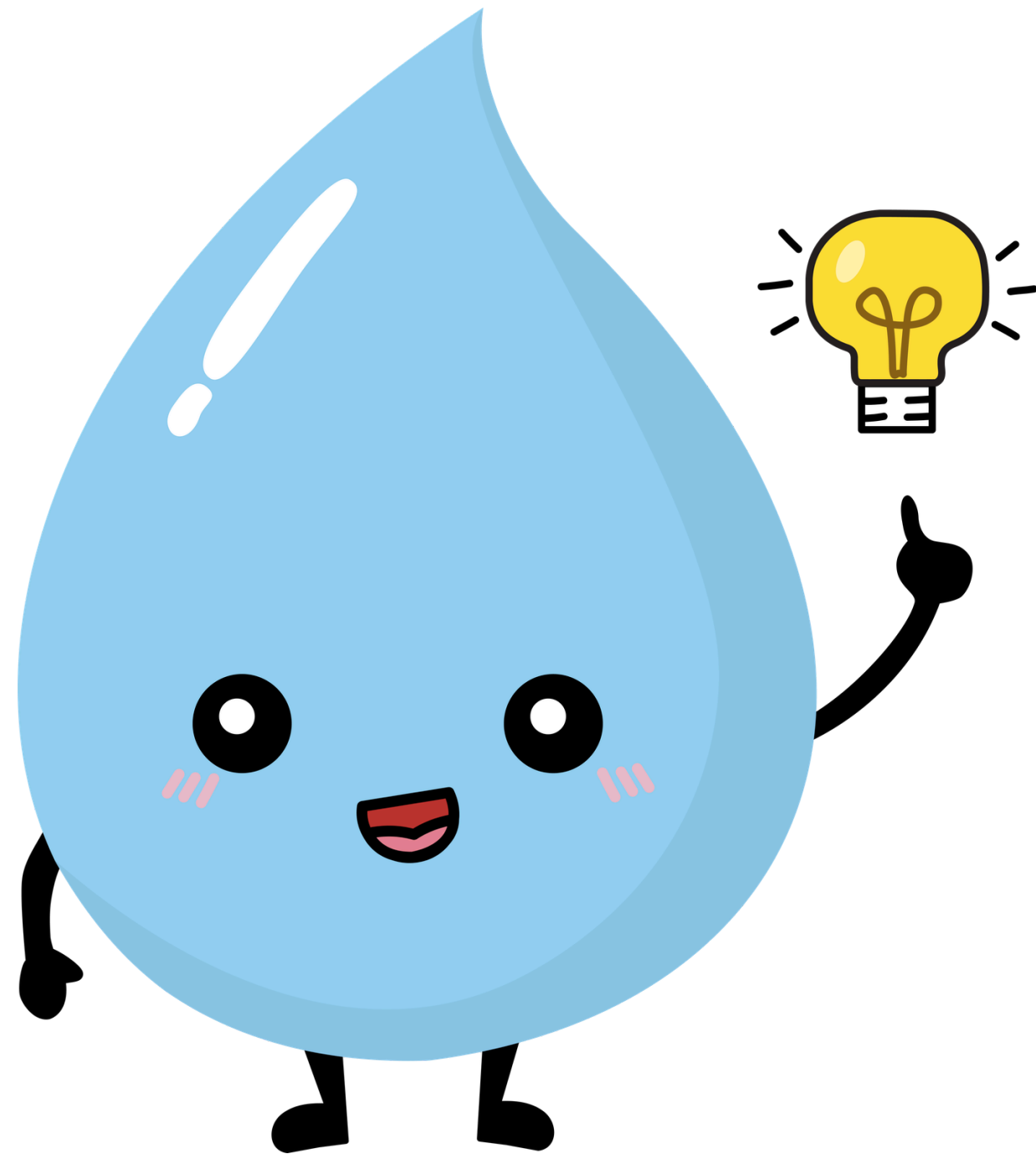
DELIVERABLES

TOOLKIT

- Includes paragraphs about how to interpret and use each tool and resource
- It provides an extra database for users with links, contacts, etc.
- **All deliverables in one place!**



INSIGHTS & KNOWLEDGE GAINED



1ST POINT

Public health initiatives are most effective when educational efforts are tied to measurable outcomes.

2ND POINT

Learned to balance depth with accessibility.

3RD POINT

Learned to design with intention, rather than just for completion.

WORKING AS A TEAM

Strong communication and flexibility allowed us to shape the project while meeting both academic and partner expectations.

● GROUP PROCESS

We supported each other by working to our strengths, setting shared goals, and keeping detailed meeting notes to stay aligned and consistent.

● PARTNER COLLABORATION

We worked closely with our partner to negotiate deliverables, adapt the project to fit our timeline, and ensure our work aligned with her vision while allowing creative input.



CHALLENGES ENCOUNTERED

Overcoming challenges required clear communication and collaborative problem solving from the whole team

● 1ST PROBLEM

The initial list of deliverables we were provided seemed very extensive and would have been difficult to complete given our limited timeframe.

● 2ND PROBLEM

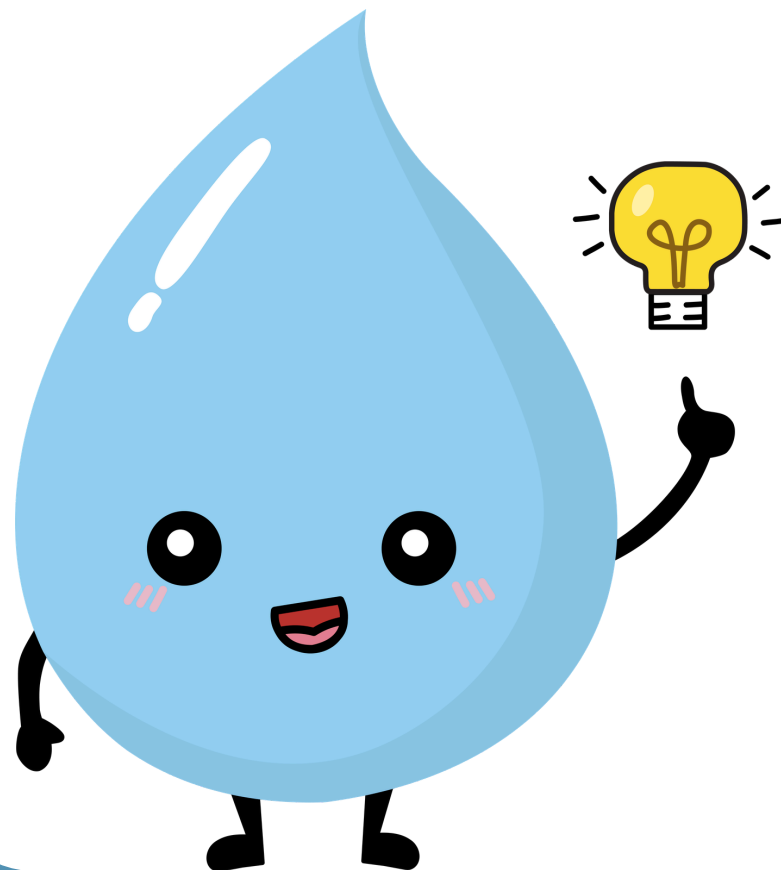
Planning meetings was difficult due to the busy schedules of the group members, as well as our facilitator.



TEAM REFLECTION 1

HOW WE WOULD PERFORM DIFFERENTLY

- Looking back on our process, we believe that the approach that was taken was very effective and efficient in completing this project.
- From the beginning of the project, communication was key, and we were able to effectively divide up the work based on strengths and set internal deadlines.
- Although minimal challenges were experienced throughout the process, we believe that they were overcome successfully as a group.
- If we were to change anything from our experience, it would be establishing more structured timelines earlier in the process



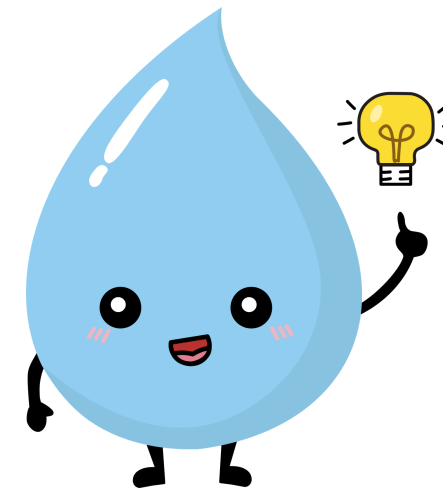
TEAM REFLECTION 2

OUR EXPERIENCES AT GLOCAL & PERSONAL GROWTH



GLOCAL EXPERIENCE

Working with GLOCAL gave us valuable insight into how community-based organizations approach real-world health challenges.



KNOWLEDGE TRANSLATION

We learned how to take complex health issues and make them more accessible through clear, engaging, and practical materials.



APPLIED LEARNING

This experience allowed us to apply course knowledge in a practical setting, making our learning more meaningful and impactful.



TEAM GROWTH

We gained confidence in my ability to contribute to real-world projects and create work that can support positive change in communities.

CONTRIBUTION & TAKEAWAYS

ASHLEY

- **ROLE:** Youth & Parent Infographics, Parent Resource Database
- **KEY TAKEAWAYS:** Raising awareness and promoting positive change doesn't have to be complex—effective knowledge translation through simple, accessible, and engaging educational materials can be highly influential. I value knowing that my work can impact individuals through education and support long-term population health.



CONTRIBUTION & TAKEAWAYS

ZSHEANA



- **ROLE:** Partner Contact, General Lead Infographic, Logic Model
- **KEY TAKEAWAYS:** Working on this project showed me how public health, research, evaluation, and problem-solving can come together to create real-world impact. I value seeing how carefully analyzing evidence and designing practical solutions can not only address health issues but also guide better decision-making and meaningful change in communities.

CONTRIBUTION & TAKEAWAYS

STEPHANIE

- **ROLE:** Parent Infographic, Impact of Issue Research Report
- **KEY TAKEAWAYS:** Working with a real-world organization showed me the importance of collaboration, adaptability, and clear communication in creating meaningful impact. I valued learning how to balance evidence-based ideas with real-world needs, while incorporating feedback to ensure solutions are practical, relevant, and aligned with organizational goals.



CONTRIBUTION & TAKEAWAYS

JADA

- **ROLE:** Stakeholder Matrix, Cartoon video
- **KEY TAKEAWAYS:** This project made me understand that not only should research be informative, but it should also be accessible and meaningful to the community it is trying to help. I learnt that substantial work goes into increasing public awareness of health issues and devising actionable steps to ensure public health.



CONTRIBUTION & TAKEAWAYS

EMMA



- **ROLE:** Toxicological Report, Epidemiological Report, Risk Assessment + Risk Management Frameworks
- **KEY TAKEAWAYS:** This project pushed me to see knowledge not as static but as something that needs to be reshaped to have a real impact. Understanding an issue is not enough, but what matters is translating it into forms that are clear, relevant, and actually usable for the community. It challenged me to think about accessibility, clarity, and how information can empower action when it is made digestible.

CONTRIBUTION & TAKEAWAYS

ASAD



- **ROLE:** Lead Detective Cartoon, Logic Model, DPSEEA Model
- **KEY TAKEAWAYS:** Working on this project reinforced the importance of adaptability while working in a group setting. Being flexible and open to feedback allowed me to not only produce high quality work, but it also helped the group ensure accountability. Furthermore, the group's work ethic helped me build habits that would aid in my ability to balance a busy workload.